Structured Pruning and Distillation under an 8 GiB GPU Budget

Executive result

Executed **18 full-size recovery conditions**, two physical compression levels, three seeds, matched CE controls, CPU/CUDA references and measured workload scaling.

Teacher: 3,085,938,688 parameters; test PPL **17.95**; PIQA200 **75.0%**. These are local measurements, not NVIDIA scores.

Scientific design

See locked plan (../EXPERIMENT_PLAN.md), config.json and exact recipes. Every main run gets 196,608 input tokens / 195,072 causal targets and 192 updates; full-vocabulary T=2 KL or ordinary CE, same rank-8 q/v LoRA budget.

The informed depth criterion is contiguous-window calibration NLL, not the earlier pilot's individual-layer ranking. Both criteria preserve boundary layers. Random architecture and training vary with seed; informed architecture is shared and training initialization/order vary.

No automatic hyperparameter sweep or test-based checkpoint selection was performed. The exported strong/informed/seed42 model is a predeclared representative, not a claimed best model.

Three-seed final comparison

Values are arithmetic mean Â± sample SD across three seeds; no significance or population-generalization claim. PPL is mean of seed PPLs, not exp(mean NLL).

Structure / criterion / objective	Parameters B	Reduction	Test PPL	PIQA accuracy
moderate/random/kl	2.501	18.96%	32.00 Â± 2.41	70.5% Â± 2.8pp
moderate/informed/kl	2.501	18.96%	38.26 Â± 0.23	72.3% Â± 0.6pp
moderate/informed/ce	2.501	18.96%	28.57 Â± 0.03	72.0% Â± 0.0pp
strong/random/kl	1.985	35.68%	72.68 Â± 13.32	60.7% Â± 2.5pp
strong/informed/kl	1.985	35.68%	69.10 Â± 0.82	65.8% Â± 0.3pp
strong/informed/ce	1.985	35.68%	52.32 Â± 0.21	65.3% Â± 0.3pp

Does minimum immediate damage predict best recovery?

Compare validation NLL before/after at the same structure and seed; downstream task ordering is a separate question. Lower NLL is better, higher PIQA is better. Do not extrapolate one criterion universally.

• **moderate, informed_minus_random_KL:** paired test Î”NLL mean +0.1808, SD 0.0829, range[+0.1176,+0.2746]; paired Î”PIQA+1.83pp. Initial validation Î”NLLs: +1.6695, -0.1955, +1.2314.

• **moderate, KL_minus_CE_informed:** paired test Î”NLL mean +0.2921, SD 0.0058, range[+0.2867,+0.2982]; paired Î”PIQA+0.33pp. Initial validation Î”NLLs: +0.0000, +0.0000, +0.0000.

• **strong, informed_minus_random_KL:** paired test Î”NLL mean -0.0397, SD 0.1680, range[-0.2218,+0.1092]; paired Î”PIQA+5.17pp. Initial validation Î”NLLs: +3.9841, -2.9802, +2.6008.

• **strong, KL_minus_CE_informed:** paired test Î”NLL mean +0.2782, SD 0.0138, range[+0.2671,+0.2936]; paired Î”PIQA+0.50pp. Initial validation Î”NLLs: +0.0000, +0.0000, +0.0000.

Direct answer at moderate compression: the criterion with lower immediate validation NLL also won final validation NLL in **2/3** seed pairs, final test NLL in **2/3**, and final PIQA in **1/3** (ties are retained explicitly). This describes these measured pairs, not a universal ranking law.

Direct answer at strong compression: the criterion with lower immediate validation NLL also won final validation NLL in **2/3** seed pairs, final test NLL in **2/3**, and final PIQA in **1/3** (ties are retained explicitly). This describes these measured pairs, not a universal ranking law.

Controlled objective result

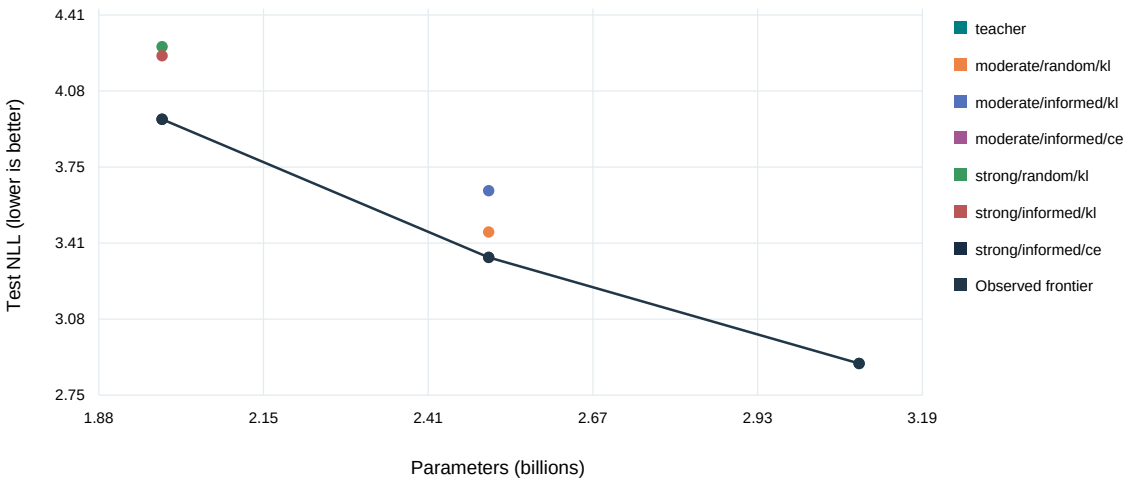
At both compression levels, the matched CE-only recovery achieved lower WikiText test perplexity than KL-only recovery at the same token budget. The measured KL-minus-CE mean PIQA effects span +0.33 to +0.50 percentage points; this small shared task set does not establish statistical superiority.

A distillation advantage must be established by measurements, not presumed. These experiments do not isolate why the objectives differ: finite token budget, adapter-only capacity, temperature and an uncorrected teacher are plausible hypotheses, not established causes. No post-hoc tuning or removal of unfavorable findings is performed.

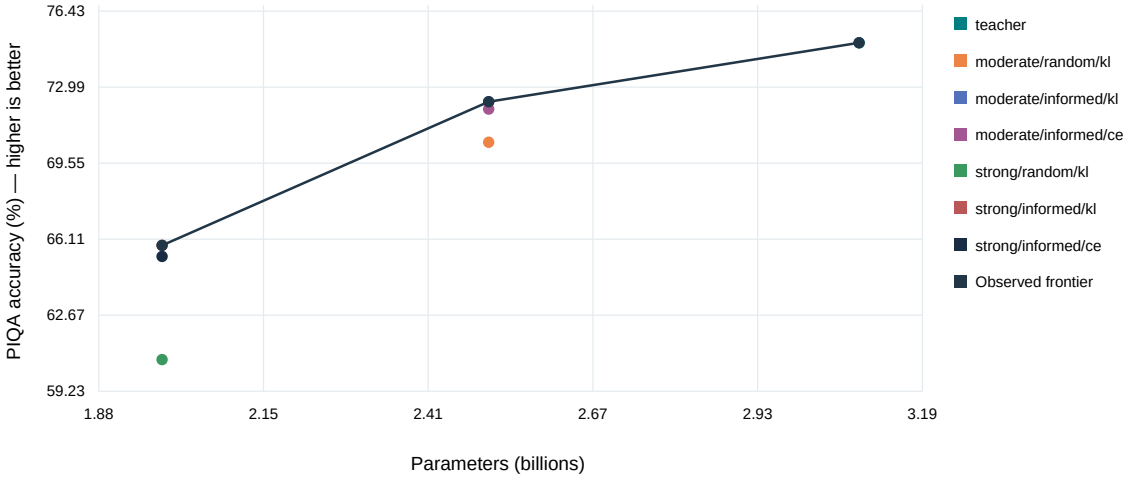
The isolated KL-minus-CE contrast changes only the objective. It does not establish NVIDIA-scale teacher-correction, full-weight training or a 50Ã— token-efficiency gain.

Equal-budget recovery (three-seed means)

Compression–quality: measured NLL frontier



Compression–quality: measured PIQA frontier



CPU / GPU execution and scientific quality

Real-model CPU FP32 versus CUDA FP16 is a **deployment device+precision comparison**, not isolated hardware acceleration. Same model recipes, rounded merged weights, held-out subset and PIQA scoring are used. CPU has 4 declared threads, leaving resources for the desktop. No fallback is allowed.

This paired quality table uses the same first **16 test blocks (2,032 causal targets) and 32 PIQA examples** on both devices. The main model table instead uses 256 blocks and 200 examples. Its PPL is therefore not numerically interchangeable with this smaller, harder subset.

Model	CPU FP32 PPL	CUDA FP16 PPL	Î”NLL GPUâˆ’CPU	CPU / GPU PIQA	Disagreements /32
teacher	24.367	24.370	+0.00010	75.0% / 75.0%	0
moderate_informed_s42_kl	52.539	52.609	+0.00134	62.5% / 62.5%	0
strong_informed_s42_kl	83.040	82.965	-0.00091	56.2% / 56.2%	0

Two warmups and five synchronized timed requests per size; raw timings and IQR are retained. The request timer separates tokenization, host/device transfer, prefill+decode and output decode. Reconstruction/loading is separate. Loading is not an OS-cold-cache measurement.

Model	Promptâ†’output	CPU model s	CUDA model s	Deployment model ratio	Request E2E ratio
teacher	32â†’16	6.096	0.582	10.48Ã—	10.44Ã—
teacher	128â†’16	7.589	0.602	12.61Ã—	12.56Ã—
teacher	512â†’16	14.822	0.668	22.19Ã—	22.13Ã—
moderate_informed_s42_kl	32â†’16	4.794	0.506	9.48Ã—	9.44Ã—
moderate_informed_s42_kl	128â†’16	5.983	0.536	11.16Ã—	11.11Ã—
moderate_informed_s42_kl	512â†’16	12.073	0.568	21.25Ã—	21.19Ã—
strong_informed_s42_kl	32â†’16	3.841	0.480	8.00Ã—	7.98Ã—

Model	Promptâ†’output	CPU model s	CUDA model s	Deployment model ratio	Request E2E ratio
strong_informed_s42_kl	128â†’16	4.689	0.466	10.05Ã—	10.02Ã—
strong_informed_s42_kl	512â†’16	9.422	0.499	18.88Ã—	18.80Ã—

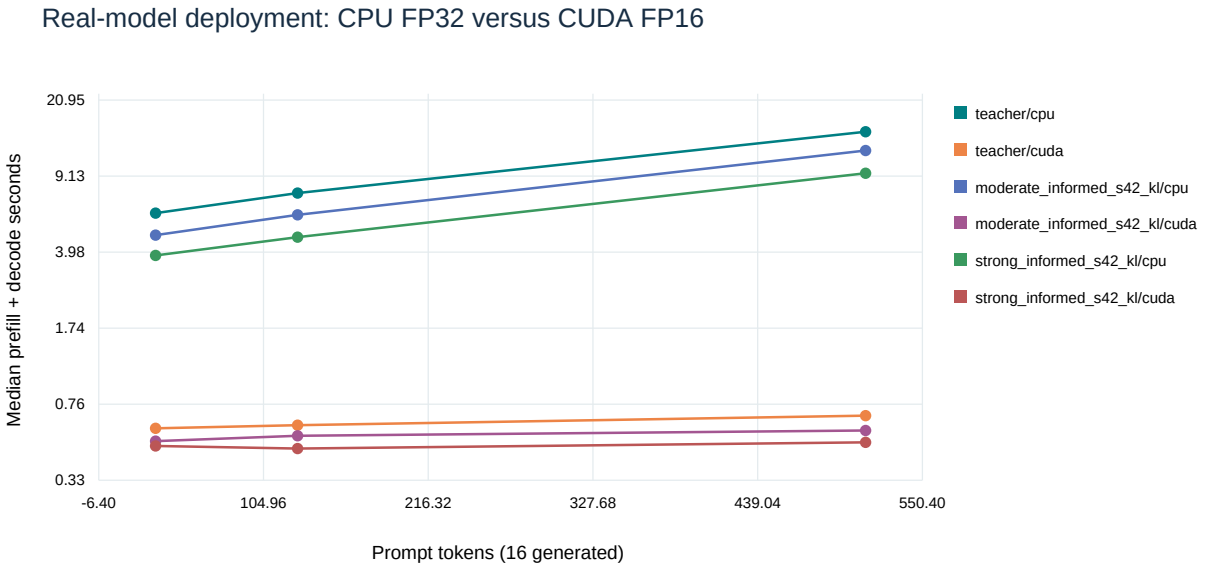
Observed benefit scale

• moderate_informed_s42_kl: CUDA deployment request benefit observed at prompt lengths [32, 128, 512]. Already beneficial at smallest tested 32 tokens; exact crossover is below/unknown relative to sampled range.

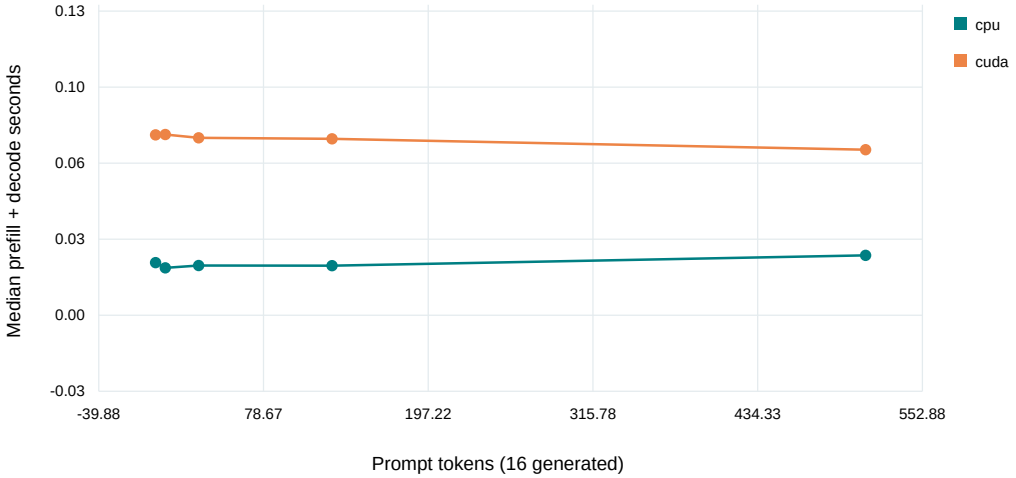
• strong_informed_s42_kl: CUDA deployment request benefit observed at prompt lengths [32, 128, 512]. Already beneficial at smallest tested 32 tokens; exact crossover is below/unknown relative to sampled range.

• teacher: CUDA deployment request benefit observed at prompt lengths [32, 128, 512]. Already beneficial at smallest tested 32 tokens; exact crossover is below/unknown relative to sampled range.

Clean hardware comparison: the tiny reference uses identical learned weights and FP32 on both devices. GPU request benefit observed at sampled lengths: none; the absence of benefit is a valid result, not a failed benchmark.



Matched FP32 tiny transformer: same weights and inputs

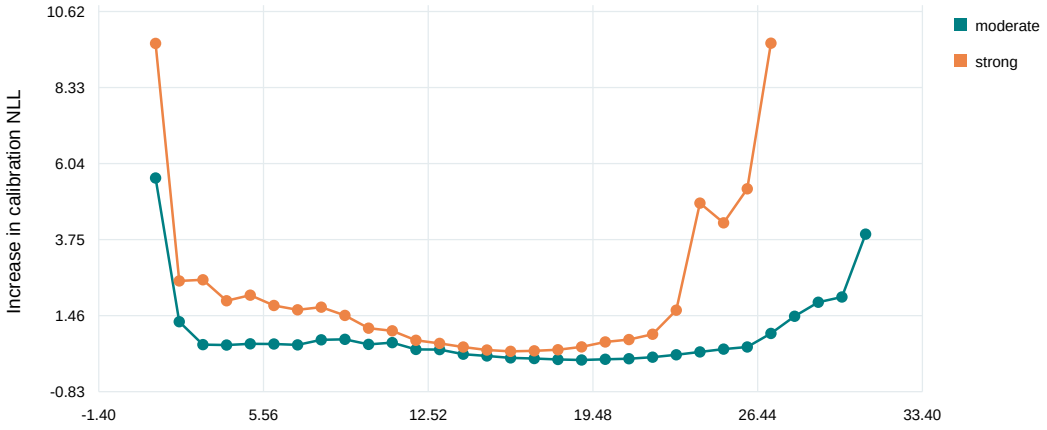


Reduced no-GPU reference

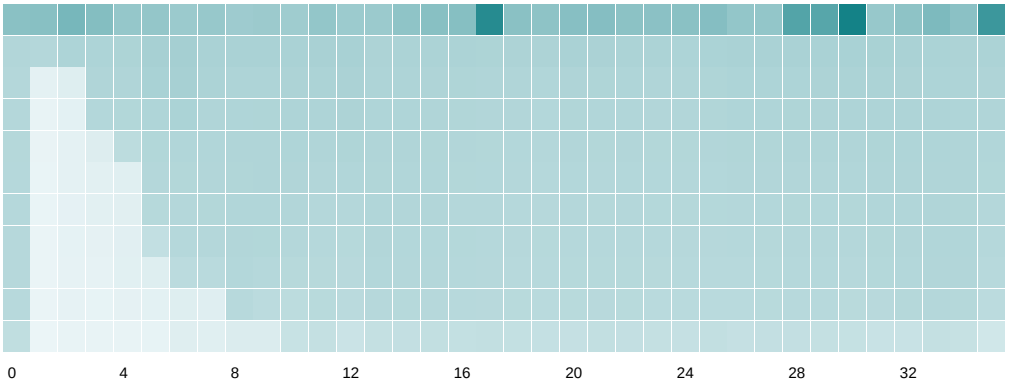
A 4-layer, 64-hidden-dimension byte-level Qwen transformer is trained on local WikiText train text; physically pruned to 3 layers / FFN 128, recovered by KL/CE on CPU and CUDA across the same three seeds. Full-weight tiny recovery is used, unlike main LoRA recovery. Complete per-seed quality, recipes, counts and timings: TINY_CPU_GPU_RESULTS.csv. Byte-level PPL must never be numerically compared to subword 3B PPL.

Sensitivity, structural removal and Pareto frontier

Contiguous depth-window sensitivity (calibration only)



FFN magnitude sensitivity distribution by layer



Original layer →; vertical axis: channel score quantiles 0% (bottom) to100% (top)

Coupled gate/up row and down column L2 energy; descriptive scores, not causal per-channel ablation.

Every recipe stores exact retained original layer IDs and coupled FFN IDs. Removed channels are the complement in 0..11007, not zeros/masks. FFN map shows magnitude distributions; only depth map is causal ablation sensitivity. The finite observed NLL Pareto frontier is: teacher, moderate/informed/ce, strong/informed/ce. Lines do not imply unmeasured architectures.

Workflow cost and resource safety

Full-vocabulary targets were streamed in 64-block shards, consumed by all conditions, then their payloads removed only after durable checkpoint receipts. Payload hashes/manifests remain. No hidden-state targets or approximate top-k objective were substituted.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    "Checkpoint timing cumulative counter excludes the most recent write due to checkpoint serialization
ordering; approximate, not exact.",
    "Power-cut/replayed steps may add wall cost; logical tokens count only committed updates.",
    "load_inclusive_first_request_seconds in raw benchmark is load plus first timed WARM request, a proxy, not
measured cold-first-request latency."
  ]
}
```


See PHASE_COSTS.json and logs/stage_costs.jsonl. Stage-wall time includes Python imports, monitoring, checkpointing and other overhead, and is not reported as estimator-only time. Main execution includes forward/backward/optimizer; target transfer is timed separately. Validation costs are not folded into optimizer time. Checkpoint-time counter is approximate as disclosed.

NVIDIA paper comparison and boundaries

• Shared principles: structural trimming, calibration ranking, random baseline, teacher-logit forward KL, token-budget recovery analysis.

• Deliberate adaptations: 3B teacher, joint depth+FFN (paper often separates depth/width), rank-8 LoRA rather than full-weight main recovery, short 128-token blocks, 196k tokens rather than 94Bâ€“380B, and no teacher correction/NAS/alignment.

• Paper reports Minitron width/depth downstream differences and teacher correction. Our measurements cannot establish those large-scale conclusions; our CE ablation only tests the local objective contrast.

• PyTorch CUDA is executed. TensorRT-LLM is conditionally excluded with official native-Windows compatibility evidence in BACKEND_SUPPORT_AUDIT.md. No TensorRT throughput claim. RAPIDS/cuML fallback: N/A.

Limitations and threats to validity

Only three seeds, two levels, one model family, one machine and a short recovery budget. Fixed calibration, custom PIQA200 and previously disclosed WikiText test partition are not broad external validation. No semantic/near-duplicate or pretraining-contamination guarantee. Exact-block/source split checks are narrower. PIQA examples and text blocks are shared across seeds; do not treat all predictions as independent replicates. Desktop load, CPU thread budget, uncontrolled OS cache and thermal state affect timings. GPU system samples are intermittent; allocator peaks are not total board memory. CPU temperature is not available from a verified sensor. The CPU tiny reference validates mechanics/trends, not robustness of 3B training on CPU. Its 512-token prompt plus 16 generated tokens exceeds the nominal 512-position configuration; execution succeeded with rotary attention, but this is not a validated long-context quality result (see PROTOCOL_AMENDMENTS.md). No API teacher or quantization claim; parameter reduction is separated from arithmetic precision. The 3B model license is noncommercial, not Apache-2.0.

Reproducibility, portfolio and evidence ledger

Reproduction commands (../REPRODUCE.md), model card (../MODEL_CARD.md), data card (../DATA_CARD.md), validation receipt (VALIDATION.json), requirements ledger (REQUIREMENTS_LEDGER.md), pinned configs/recipes, all 18 final adapters and raw per-seed/per-example JSON. Saved optimizer checkpoints and STOP/RESUME commands tolerate normal interruption but cannot protect against all disk/power faults. The sealed earlier submission is preserved as the pilot, not overwritten.

References

Sreenivas et al. (2024), LLM Pruning and Distillation in Practice: The Minitron Approach, supplied technical report; https://arxiv.org/abs/2408.11796 .

Muralidharan et al. (2024), Compact Language Models via Pruning and Knowledge Distillation; https://arxiv.org/abs/2407.14679 .

Qwen2.5-3B model card/license: https://huggingface.co/Qwen/Qwen2.5-3B .

WikiText: https://huggingface.co/datasets/Salesforce/wikitext . PIQA: https://github.com/ybisk/ybisk.github.io/tree/master/piqa .

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